

Research Interests	Natural language processing, reasoning, interpretability, parameter- and data-efficient methods, and synthetic data generation.	
Education	University of Illinois Urbana-Champaign, IL, USA <i>Ph.D. in Electrical and Computer Engineering</i>	Expected Dec 2026
	Indian Institute of Technology Roorkee, Uttarakhand, India <i>B.Tech. in Electronics and Communication Engineering</i> <i>M.Tech. in Wireless Communication</i>	2009 - 2014
Employment	University of Illinois Urbana-Champaign, IL, USA <i>Graduate Research Assistant</i>	2021 - Present
	I am currently studying LLMs for mathematical reasoning. Previously, I worked on representing mathematical expressions in a continuous vector space (similar to Word2Vec for natural text) and the auto-formalization of linear programming problems.	
	Myntra, Bengaluru, KA, India <i>Technical Lead</i>	2019 - 2021
	I led a team of three people that managed inventory systems at Myntra. I was responsible for maintaining and supporting the inventory systems as well as the software architecture and design of new features. I was also responsible for building the technical roadmap for the team. The projects included scaling inventory systems to support 1000x traffic and salvaging orphan inventory in Myntra warehouses.	
	Senior Software Engineer	2017 - 2019
	I worked in the tech team for warehouse management. The main projects included last-mile consolidation and multi-tenancy to enable multiple tenants on the existing systems. I also led a team of four people to overhaul the design and architecture of the stock transfer workflow. I acquired skills in system design and architecture, asynchronous frameworks, and team management.	
Internships	AIANash/Shoplane, Bengaluru, KA, India <i>Co-founder</i>	2015 - 2017
	I worked in a team of three on a platform to improve online customer experience by predicting user intent. The initial idea was to develop a smart shopping experience employing artificial intelligence. It later shifted to creating a B2B artificial intelligence-driven analytics platform.	
	Commonfloor, Bengaluru, KA, India <i>Software Development Engineer 1</i>	2014 - 2015
	I was part of the revenue team in the real estate vertical. I worked on the lead sharing and pricing modules. I also developed a JavaScript plugin to capture impressions and scripts to report ad performance on product pages.	
	Amazon, Palo Alto, CA, USA <i>Applied Scientist Intern</i>	Summer 2025
	I worked on parameter-efficient fine-tuning methods.	
	Amazon, Palo Alto, CA, USA <i>Applied Scientist Intern</i>	Summer 2024
	I worked on parameter-efficient multi-task learning.	

	IBM India Software Labs , Bengaluru, KA, India <i>Extreme Blue Internship Program</i> Summer 2013 I worked on the timing analysis of digital circuits and created core data models for GLSM that were modular and multi-threadable.
	Defence Research and Development Organisation , Bengaluru, KA, India <i>Summer Intern</i> Summer 2012 I analyzed the helix slow-wave structure (SWS) and compared the results with HFSS results.
Relevant Coursework	Pattern Recognition, Mathematical Models of Language, Efficiency in NLP, Deep Learning for Computer Vision, Computer Vision, Probability Theory, Information and Communication Theory, Adaptive Filter Theory, Data Structures and Algorithms.
Awards	1st Place, NL4Opt competition (subtask 2) at NeurIPS 2022 (nl4opt.github.io) 2022
Publications	6. GIVA: Gradient-Informed Bases for Vector-Based Adaptation Neeraj Gangwar, Rishabh Deshmukh, Michael Shavlovsky, Hanco Li, Vivek Mittal, Lexing Ying, Nickvash Kani <i>International Conference on Artificial Intelligence and Statistics</i>, 2026 (To Appear; Spotlight, top 2.5%) 5. Parameter-Efficient Multi-Task Learning via Progressive Task-Specific Adaptation Neeraj Gangwar, Anshuka Rangi, Rishabh Deshmukh, Holakou Rahmanian, Yesh Dattatreya, Nickvash Kani <i>International Conference on Artificial Intelligence and Statistics</i>, 2026 (To Appear) [arXiv] 4. E-Gen: Leveraging E-Graphs to Improve Continuous Representations of Symbolic Expressions Hongbo Zheng*, Suyuan Wang*, Neeraj Gangwar, Nickvash Kani <i>Nations of the Americas Chapter of the Association for Computational Linguistics</i>, 2025 [ACL Anthology] *Equal contribution 3. Integrating Arithmetic Learning Improves Mathematical Reasoning in Smaller Models Neeraj Gangwar, Suma Bhat, Nickvash Kani <i>Language Resources and Evaluation Conference</i>, 2026 (To Appear) [arXiv] <i>Math-AI Workshop at NeurIPS</i>, 2024 [OpenReview] 2. Semantic Representations of Mathematical Expressions in a Continuous Vector Space Neeraj Gangwar, Nickvash Kani <i>Transactions on Machine Learning Research</i>, 2023 [OpenReview, arXiv] 1. Highlighting Named Entities in Input for Auto-Formulation of Optimization Problems Neeraj Gangwar, Nickvash Kani <i>International Conference on Intelligent Computer Mathematics</i>, 2023 [Springer, arXiv]
Technical Skills	Python, PyTorch, Java, MySQL, Cassandra, Kafka, RabbitMQ, Spring, Hibernate, Redis.
Teaching	ECE364: Programming Methods for Machine Learning (Spring 2025, Spring 2026) University of Illinois Urbana-Champaign Teaching Assistant with Prof. Nickvash Kani ECE374B: Introduction of Algorithms and Models of Computation (Fall 2024, Fall 2025) University of Illinois Urbana-Champaign Teaching Assistant with Prof. Nickvash Kani EC362: Communication Systems Lab (Autumn 2013) Indian Institute of Technology Roorkee Teaching Assistant with Prof. Debashis Ghosh

Extra-Curricular Activities	Student Mentor at DIYA Research (diya-research.org)	2022
	Member of SDS Labs (sdslabs.co), IIT Roorkee	2010 - 2014
	Coordinator of the Linux Group, IIT Roorkee	2011 - 2012
	Member of the National Service Scheme (NSS), IIT Roorkee	2009 - 2010